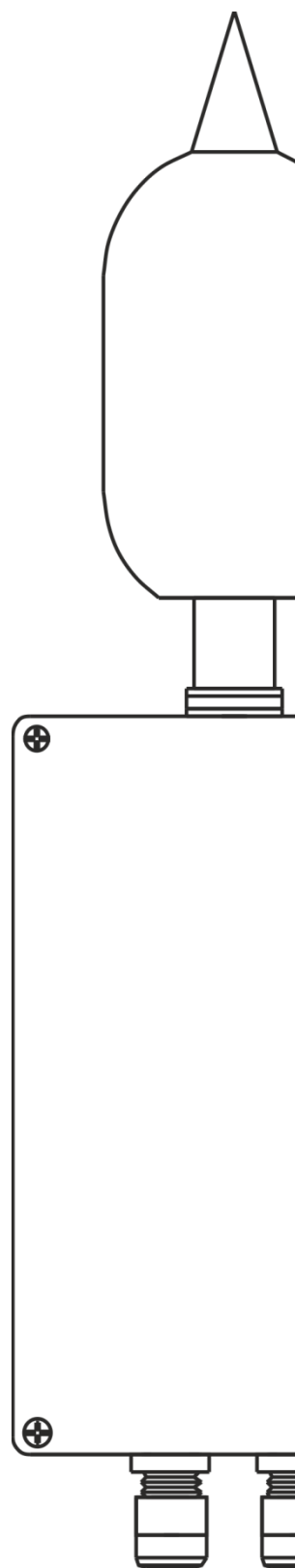


TA150

COMMUNICATION PROTOCOLS

PR_TA150_v0007_20240729_EN



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1.1 Sensor configuration (webserver)

To configure the sensor, you must enter the sensor's webserver. Access to the webserver is done with an external device that has a web browser and WIFI connection: desktop or laptop computer, tablet, smartphone, etc.

The access to the webserver must be done by activating the direct mode of the TA150, to do this:

- Press once on the push button to activate the display.
- Keep the button pressed until the bar that appears on the display is complete.
- After the bar is completed, the sensor activates the direct mode of communication by generating a WIFI wireless communication.

Communication ends when you press the button again until the bar is complete on the display, or after 60 minutes of inactivity.

To communicate the desired device (PC, tablet, smartphone ...) with the TA150 sensor follow the next steps:

- Search for WIFI networks with the device you want to connect to the TA150 sensor.
- Select the WIFI network that generates the TA150 sensor (displayed on the sensor display).
- Connect the device to the WIFI network and enter the required password (displayed on the sensor display). If the operating system reports that this network does not allow access to the Internet, indicate that you want to maintain the connection.
- Finally open an Internet browser and write in the address bar the webserver's IP address (displayed on the sensor display).

Now you can access the sensor configuration by browsing the webserver pages. For more information, please refer to the sensor manual.

1.2 Function's names table

The TA150 register the follow functions:

Overall functions:

Function	Level	Overload	Underrange
LAT	la	la-o	la-u
LCT	lc	lc-o	lc-u
LZT	lz	lz-o	lz-u
LAIT	lai	lai-o	lai-u
LAFmaxT	lafmx	lafmx-o	lafmx-u
LASmaxT	lasmx	lasmx-o	lasmx-u

Function	Level	Overload	Underrange
LAlmaxT	laimx	laimx-o	laimx-u
LCpeakT	lcpk	lcpk-o	lcpk-u
LZpeakT	lzpck	lzpck-o	lzpck-u
L10	L10	L10-o	L10-u
L50	L50	L50-o	L50-u
L90	L90	L90-o	L90-u

Spectrum (1/3 octave) functions (only available if the **FR150** module is installed):

Function	Level	Overload	Underrange
6 Hz	l006	l006-o	l006-u
8 Hz	l008	l008-o	l008-u
10 Hz	l010	l010-o	l010-u
12.5 Hz	l012	l012-o	l012-u
16 Hz	l016	l016-o	l016-u
20 Hz	l020	l020-o	l020-u
25 Hz	l025	l025-o	l025-u
31.5 Hz	l031	l031-o	l031-u
40 Hz	l040	l040-o	l040-u
50 Hz	l050	l050-o	l050-u
63 Hz	l063	l063-o	l063-u
80 Hz	l080	l080-o	l080-u
100 Hz	l100	l100-o	l100-u
125 Hz	l125	l125-o	l125-u
160 Hz	l160	l160-o	l160-u
200 Hz	l200	l200-o	l200-u
250 Hz	l250	l250-o	l250-u
315 Hz	l315	l315-o	l315-u

Function	Level	Overload	Underrange
400 Hz	l400	l400-o	l400-u
500 Hz	l500	l500-o	l500-u
630 Hz	l630	l630-o	l630-u
800 Hz	l800	l800-o	l800-u
1000 Hz	l1k0	l1k0-o	l1k0-u
1250 Hz	l1k2	l1k2-o	l1k2-u
1600 Hz	l1k6	l1k6-o	l1k6-u
2000 Hz	l2k0	l2k0-o	l2k0-u
2500 Hz	l2k5	l2k5-o	l2k5-u
3150 Hz	l3k1	l3k1-o	l3k1-u
4000 Hz	l4k0	l4k0-o	l4k0-u
5000 Hz	l5k0	l5k0-o	l5k0-u
6300 Hz	l6k3	l6k3-o	l6k3-u
8000 Hz	l8k0	l8k0-o	l8k0-u
10000 Hz	l10k	l10k-o	l10k-u
12500 Hz	l12k	l12k-o	l12k-u
16000 Hz	l16k	l16k-o	l16k-u
20000 Hz	l20k	l20k-o	l20k-u

1.3 Online command table

The TA150 accepts the follow commands:

Command name	Command value	Format value	Description
tl	0001 to 3600	Number (4 digits)	Averaging time parameter in seconds default is 0060
fa-ou	1 or 0 true or false	Boolean	Deactivate or activate the “L _{AT} function Overload/Underrange” option
fc fz fai fafmx fasmx faimx fcpk fzpk f10 f50 f90	1 or 0 true or false	Boolean	Deactivate or activate the “functions registers” option
fex-ou	1 or 0 true or false	Boolean	Deactivate or activate the “functions overload/Underrange” option
f1s	1 or 0 true or false	Boolean	(*2) Activate or deactivate the “1s registers” option
f1s_n	1 ... 9	Number	(*2) Selects the 1s registers' function: 1 = LA 2 = LC 3 = LZ 4 = LAI 5 = LAFmax 6 = LASmax 7 = LAImax 8 = LCpeak 9 = LZpeak
f1s-ou	1 or 0 true or false	Boolean	(*1) Deactivate or activate the “1s reg. Overload/Underrange” option
fto	1 or 0 true or false	Boolean	(*1) Deactivate or activate the “1/3 registers” option
fto-ou	1 or 0 true or false	Boolean	(*1) Deactivate or activate the “1/3 Overload/Underrange” option

*1: If **FR150** module is installed

*2: If t ≤ 60s

1.4 Seeing the server-device communication

After the device is configured, you can view the HTTP/HTTPS communication between server and device through the USB interface.

To do this, you need to connect the device to a PC with a USB-B cable and have an emulator terminal installed (for example, Tera Term). After connection, you may check the "Serial" option and select the USB port.

Once communication is established, you can use two commands:

- **Command with letter "X"** (upper case): allows you to view the HTTP/HTTPS shipments from the device and the HTTP/HTTPS responses from the server. Remember that this information will be displayed based on the integration time configured in the sensor (TL parameter).
- **Command with letter "x"** (lower case): deactivates the display of "X".

PLATFORM REGISTERS PROTOCOLS

2

2.1 UltraLight 2.0 (UL2.0)

2.1.1 Sending data

Format	Protocol	Method
UL2.0 DEVICE API	HTTP, HTTPS MQTT-TCP, MQTT-TLS	POST

Header description:

```
POST/<path>?k=<key>&i=TA150-<serialnum>&t=<time>&getCmd=<gc> HTTP/1.1
Host: <host>
Content-Type: text/plain; charset=UTF-8
```

URL	Name		Description
	host		(*1) Server API domain name or IP
		path	(*1) Server API path
Parameters	Name	Value	Description
	k	key	(*1) server API Key
	i	serialnum	Device serial number (Txxxxxx)
	t	time	Timestamp (in Zulu format: yyyy-mm-ddThh:ii:ssZ)
	getCmd	gc	request for receive commands (boolean: 1 = yes -default-, 0 = no)

*1: Programmable by webserver

Body description:

Generic line structure (separated by “|” character):

```
<name1>|<value1>|<name2>|<value2> ... |<nameX>|<valueX>
```

There are 5 parts in the body structure:

Overall values:

```
t1|<t1>|la|<la>|la-o|<la-o>|la-u|<la-u>| ... |l90|<l90>|l90-o|<l90-o>|l90-u|<l90-u>
```

Name	Description
T1	Averaging time in seconds (s)
la ... l90	Level function's sound pressure level in dB (see section 1.2)
la-o ... l90-o	(*2) Level function's Overload [boolean: 1 = true, 0 = false] (see section 1.2)
la-u ... l90-u	(*2) Level function's Underrange [boolean: 1 = true, 0 = false] (see section 1.2)

*2: If “Send Overload/Underrange” is activated (programmable by Webserver)

Seconds values:

```
|f1s_n|<1sf>
|11s|<f01s>;<f02s>; ... ;<f_xxs>
|11s-o|<f01s-o>;<f02s-o>; ... ;<f_xxs-o>
|11s-u|<f01s-u>;<f02s-u>; ... ;<f_xxs-u>
```

Name	Description
f1s_n	(*3) Level function's 1s name (see section 1.2)
1s	(*3) Seconds value separated by ";" [register format: value]
11s-o	(*3) (*5) Overload values separated by ";" [register format: boolean: true or false]
11s-u	(*3) (*5) Underrange values separated by ";" [register format: boolean: true or false]

*3: If TLeq=<60s, "Send 1s registers" activated and a "1s function" selected (programmable by Webserver)

*5: If "Send 1s Overload/Underrange" is activated (programmable by Webserver)

Spectrum values: (optional, needs FR150 module)

```
|1006|<1006>|1006-o|<1006-o>|1006-u|<1006-u>| ... |120k|<120k>|120k-o|<120k-o>|120k-u|<120k-u>
```

Name	Description
I006 ... I20k	(*4) 1/3 Octave filter's sound pressure level (see section 1.2)
I006-o ... I20k-o	(*4) (*5) 1/3 Octave filter's Overload [boolean: 1 = true, 0 = false] (see section 1.2)
I006-u ... I20k-u	(*4) (*5) 1/3 Octave filter's Underrange [boolean: 1 = true, 0 = false] (see section 1.2)

*4: If optional FR150 module is installed and activated (programmable by Webserver)

*5: If "Send 1/3 Overload/Underrange" is activated (programmable by Webserver)

System values:

```
|<name1>|<value1>|<name2>|<value2> ... |<nameX>|<valueX>
```

Name	Description
pwr	Power supply status [1 = 230V, 2 = battery, 3 = 12V, 4 = POE]
wifi	(*6) Wi-fi strength (%)
bat	(*7) Battery level (%)
modem	Optional module MR154 OR MR151 - Modem strength (%)
gps	Optional module MR154 OR MR151 - Sensor coordinates in latitude and longitude

*6: If transmission by WiFi is activated

*7: If a BA150 or BA157 battery is connected to sensor

2.1.2 Response data

The server will respond with an HTTP/HTTPS status code “**200**” if the request was accepted and processed correctly.

Other code different to “200”, the sensor will consider a server’s error and will save the register for the next connection.

2.1.3 Sending commands o orders

The sensor can receive commands/orders to **change the configuration of sending registers data**. These commands can be received in the body of the server response.

If there is **a command** or **various commands** to change the configuration, the server must respond with the following format within the body of the response:

```
TA150-<serialnum>@setConfig|<cmd_name>=<cmd_value>  
HTTP/1.1
```

Parameter	Description
serialnum	Device serial number (Txxxxxx)
cmd_name	Command name
cmd_value	Command value

The TA150 accepts a lot of prebuilt online commands.

Please see **section 1.3** for those online commands.

2.1.4 Example

- Optional modules: BA150 (battery) and MR150 (4G modem) - Serial number sensor: T123456 - URL: yourserver.com/sensor/data/ - Port: 80 - Token: 98765abcdefgh	- Integration time: 60s - "Send Overload/Underrange" activated - "Send 1s registers" activated, and "LAFmaxT function" selected - "Send 1/3 registers" activated - "Send 1/3 Overload/Underrange" activated
---	---

The sensor data:

```
POST /sensor/data/?k=98765abcdefg&i=TA150-T123456&t=2028-06-10T14:12:14Z&getCmd=1
Host: yourserver.com:0080
Content-Type: None
Content-Type: text/plain; charset=UTF-8
Content-Length: 1910
Connection: close
```

[illegible]

When there isn't a command to change, the server will respond the following data:

```
HTTP/1.1 200 OK
Date: Thu, 10 Jun 2028 14:13:38 GMT
Content-Type: none
Content-Length: 0
Connection: close
```

And if there is a command (f.e. change the averaging time to **30 seconds**), the server will respond the following data:

```
HTTP/1.1 200 OK
Date: Thu, 10 Jun 2028 14:13:38 GMT
Content-Type: text/plain; charset=UTF-8
Content-Length: 32
Connection: close
```

TA150-T123456@setConfig|t1=0030

2.2 Generic JSON Format

2.2.1 Sending data

Format	Protocol	Method
JSON	HTTP, HTTPS MQTT-TCP, MQTT-TLS	PUT

Header description:

```
PUT/<path> HTTP/1.1
Host: <host>
API-KEY: <token>
Content-Type: application/json; charset=UTF-8
```

Parameters	Name	Description
	host	(*1) Server API domain name or IP
	path	(*1) Server API path
	token	(*1) Server API key secure

*1: Programmable by webserver

Body description:

There are 5 parts in the JSON structure:

```
{
  [information values]
  "overall": {
    [Overall values]
  },
  "system": {
    [system values]
  },
  "seconds": {
    [seconds values]
  },
  "spectrum": {
    [spectrum values]
  }
}
```

Information values:

```
"name": "TA150-<serialnum>",  
"manufacturer": "CESVA",  
"model": "TA150",  
"serial": "<serialnum>",  
"timestamp": "<time>",  
"averagingtime": <t1>,
```

Name	Description
serialnum	Device serial number (Txxxxxx)
time	Timestamp (in Zulu format: yyyy-mm-ddThh:ii:ssZ)
t1	Averaging time in seconds (s)

Overall values:

```
"la": <la>, "la-o": <la-o>, "la-u": <la-u>,  
[ .. ]  
"l90": <l90>, "l90-o": <l90-o>, "l90-u": <l90-u>
```

Name	Description
la ... l90	Level function's sound pressure level in dB (see section 1.2)
la-o ... l90-o	(*2) Level function's Overload [boolean: true, false] (see section 1.2)
la-u ... l90-u	(*2) Level function's Underrange [boolean: true, false] (see section 1.2)

*2: If "Send Overload/Underrange" is activated (programmable by Webserver)

System values:

```
"<name1>": <value1>,  
"<name2>": <value2>,  
[ .. ]  
"<nameX>": <valueX>
```

Name	Description
pwr	Power supply status [1 = 230V, 2 = battery, 3 = 12V, 4 = POE]
Wifi	(*6) Wi-fi strength (%)
bat	(*7) Battery level (%)
modem	(*8) Optional module MR154 or MR151 - Modem strength (%)
gps	(*8) Optional module MR154 OR MR151 - Sensor coordinates in latitude and longitude

*6: If transmission by WiFi is activated

*7: If a BA150 or BA157 battery is connected to sensor

*8: If transmission by modem is activated

Seconds values:

```
"f1s_n": "<1sf>",
"l1s" : "<f01s>;<f02s>; ... ;<f_xxs>",
"l1s-o": "<f01s-o>;<f02s-o>; ... ;<f_xxs-o>",
"l1s-u": "<f01s-u>;<f02s-u>; ... ;<f_xxs-u>"
```

Name	Description
f1s_n	(*3) Level function's 1s name (see section 1.2)
1s	(*3) Seconds value separated by “;” [register format: value]
l1s-o	(*3) (*5) Overload values separated by “;” [register format: boolean: true or false]
l1s-u	(*3) (*5) Underrange values separated by “;” [register format: boolean: true or false]

*3: If TLeq=<60s, “Send 1s registers” activated and a “1s function” selected (programmable by Webserver)

*5: If “Send 1s Overload/Underrange” is activated (programmable by Webserver)

Optional spectrum values:

```
"1006": <1006>, "1006-o": <1006-o>, "1006-u": <1006-u>,
[ .. ]
"120k": <120k>, "120k-o": <120k-o>, "120k-u": <120k-u>
```

Name	Description
l006 ... l20k	(*4) 1/3 Octave filter's sound pressure level (see section 1.2)
l006-o ... l20k-o	(*4) (*5) 1/3 Octave filter's Overload [boolean: 1 = true, 0 = false] (see section 1.2)
l006-u ... l20k-u	(*4) (*5) 1/3 Octave filter's Underrange [boolean: 1 = true, 0 = false] (see section 1.2)

*4: If optional FR150 module is installed and activated (programmable by Webserver)

*5: If “Send 1/3 Overload/Underrange” is activated (programmable by Webserver)

2.2.2 Response data

The server will respond with an HTTP/HTTPS status code “**200**” if the request was accepted and processed correctly.

Other code different to “200”, the sensor will consider a server’s error and will save the register for the next connection.

2.2.3 Sending commands o orders

The sensor can receive commands/orders to **change the configuration of sending registers data**. These commands can be received in the body of the server response.

If there is a **command** or **various commands** to change the configuration, the server must respond with the following format within the body of the response:

```
{
  "setConfig": {
    "<cmd_name2>": "<cmd_value1>",
    [..]
    "<cmd_nameX>": "<cmd_valueX>"
  }
}
```

Parameter	Description
cmd_name	Command name
cmd_value	Command value

The TA150 accepts a lot of prebuilt online commands.

Please see **section 1.3** for those online commands.

2.2.4 Example

- Optional modules: BA150 (battery) and MR150 (4G modem) - Serial number sensor: T123456 - URL: yourserver.com/sensor/data/ - Port: 80 - Token: 98765abcdefg	- Integration time: 60s - "Send Overload/Underrange" activated - "Send 1s registers" activated, and "Lz function" + "Overload/Underrange" selected - "Send 1/3 registers" activated - "Send 1/3 Overload/Underrange" activated -
--	--

```
POST /sensor/data/
Host: yourserver.com:0080
Content-Type: text/plain; charset=UTF-8
Connection: close
{
  "name": "TA150-T123456",
  "manufacturer": "CESVA",
  "model": "TA150",
  "serial": "T123456",
  "timestamp": "2028-06-10T14:12:14Z",
  "averagingtime": 60,
  "overall": {
    "la": 63.2, "la-o": false, "la-u": false,
    "lc": 63.2, "lc-o": false, "lc-u": false,
    "lz": 63.2, "lz-o": false, "lz-u": false,
    "lai": 63.2, "lai-o": false, "lai-u": false,
    "lafmx": 63.2, "lafmx-o": false, "lafmx-u": false,
    "lasmx": 63.2, "lasmx-o": false, "lasmx-u": false,
    "laimx": 63.2, "laimx-o": false, "laimx-u": false,
    "lcpk": 63.2, "lcpk-o": false, "lcpk-u": false,
    "lzp": 63.2, "lzp-o": false, "lzp-u": false,
    "l10": 63.2, "l10-o": false, "l10-u": false,
    "l150": 63.2, "l150-o": false, "l150-u": false,
    "l190": 63.2, "l190-o": false, "l190-u": false
  },
  "system": {
    "pwr": 1,
    "bat": 78,
    "modem": 89,
    "gps": "41.44195289443725, 2.2036654046947652"
  },
  "seconds": {
    "fls_n": "lz",
    "l1s": "63.2;63.2;63.2;63.2;63.2; [...] 63.2;63.2;63.2;63.2;63.2",
    "l1s-o": "false;false;false;false;false; [...] false;false;false;false;false",
    "l1s-u": "false;false;false;false;false; [...] false;false;false;false;false"
  },
  "spectrum": {
    "l006": 59.1, "l006-o": false, "l006-u": false,
    "l008": 59.1, "l008-o": false, "l008-u": false,
    "l010": 59.1, "l010-o": false, "l010-u": false,
    "l012": 59.1, "l012-o": false, "l012-u": false,
    [...],
    "l10k": 59.1, "l10k-o": false, "l10k-u": false,
    "l12k": 59.1, "l12k-o": false, "l12k-u": false,
    "l16k": 59.1, "l16k-o": false, "l16k-u": false,
    "l20k": 59.1, "l20k-o": false, "l20k-u": false
  }
}
```

When there isn't a command to change, the server will respond the following data:

```
HTTP/1.1 200 OK
Date: Thu, 10 Jun 2028 14:13:38 GMT
Content-Type: none
Content-Length: 0
Connection: close
```

And if there is a command (f.e. change the **averaging time** to **30 seconds** and **deactivate seconds**), the server will respond the following data:

```
HTTP/1.1 200 OK
Date: Thu, 10 Jun 2028 14:13:38 GMT
Content-Type: text/plain; charset=UTF-8
Content-Length: 32
Connection: close
{
  "setConfig": {
    "tl": 30,
    "ls": 0
  }
}
```

2.3 Sentilo API

2.3.1 Sending data

Format	Protocol	Method
JSON	HTTP, HTTPS MQTT-TCP, MQTT-TLS	PUT

Header description:

```
PUT/data/<provider_id> HTTP/1.1
Host: <host>:<port>
IDENTITY_KEY: <token>
Content-Type: application/json; charset=UTF-8
```

Parameters	Name	Description
	provider_id	(*1) Sentilo provider ID
	host	(*1) Server API domain name or IP
	port	(*1) Server API port in 4 digits
	token	(*1) Server API key secure

*1: Programmable by Webserver

Body description:

Basic format standar:

```
{ "sensors": [
  {
    "sensor": "TA150-<serialnum>-<name1>",
    "observations": [{"value": "<value1>", "timestamp": "<time>"}]
  }, {
    "sensor": "TA150-<serialnum>-<name2>",
    "observations": [{"value": "<value2>", "timestamp": "<time>"}]
  }, {
    ...
  }, {
    "sensor": "TA150-<serialnum>-<nameX>",
    "observations": [{"value": "<valueX>", "timestamp": "<time>"}]
  }
]}
```

	Name	Value description
Id values	serialnum	Device serial number (Txxxxxx)
	time	Timestamp (in UTC format: dd/mm/yyyyThh:ii:ssUTC)
	nameX	Sensor name's parameter
	valueX	Sensor value's parameter

<pre> { "sensor": "TA150-<serialnum>-tl", "observations": [{"value": "<tl value>", "timestamp": "<time>"}] }, { "sensor": "TA150-<serialnum>-la", "observations": [{"value": "<la value>", "timestamp": "<time>"}] }, { "sensor": "TA150-<serialnum>-la-o", "observations": [{"value": "<la-o value>", "timestamp": "<time>"}] }, { "sensor": "TA150-<serialnum>-la-u", "observations": [{"value": "<la-u value>", "timestamp": "<time>"}] }, { ... }, { "sensor": "TA150-<serialnum>-pwr", "observations": [{"value": "<pwr value>", "timestamp": "<time>"}] }, { ... } </pre>		
	Name	Value description
Overall values	la ... lzpk	Level function's sound pressure level in dB (see section 1.2)
	la-o ... lzpk-o	(*2) Level function's Overload [boolean]
	la-u ... lzpk-u	(*2) Level function's Underrange [boolean]
System values	tl	Averaging time in seconds (s)
	pwr	Power supply status [1 = 230V, 2 = battery, 3 = 12V, 4 = POE]
	wifi	(*6) Wi-fi strength (%)
	bat	(*7) Battery level (%)
	modem	Optional MR154 OR MR151 module - Modem strength (%)
	gps	Optional MR154 OR MR151 module - Sensor coordinates in latitude and longitude

*2: If "Send Overload/Underrange" is activated (programmable by Webserver)

*6: If transmission by WiFi is activated

*7: If a BA150 or BA157 battery is connected to sensor

<pre> { "sensor": "TA150-<serialnum>-f1s_n", "observations": [{"value": "<f1s value>", "timestamp": "<time>"}] }, { "sensor": "TA150-<serialnum>-l1s", "observations": [{"value": "<S1 value>;<S2 value>;<S3 value>;...;<Sxx value>", "timestamp": "<time>"}] }, { "sensor": "TA150-<serialnum>-l1s-o", "observations": [{"value": "<S1-o value>;<S2-o value>;<S3-o value>;...;<Sxx-o value>", "timestamp": "<time>"}] }, { "sensor": "TA150-<serialnum>-l1s-u", "observations": [{"value": "<S1-u value>;<S2-u value>;<S3-u value>;...;<Sxx-u value>", "timestamp": "<time>"}] } </pre>		
	Name	Value description
Seconds pack values	f1s_n	(*3) Level function's 1s name (see section 1.2)
	l1s	(*3) Seconds values separated by ";" [register format: value]
	l1s-o	(*3) (*5) Overload values separated by ";" [register format: boolean (true or false)]
	l1s-u	(*3) (*5) Underrange values separated by ";" [register format: boolean (true or false)]

*3: If TLeq=<60s, "Send 1s registers" activated and a "1s function" selected (programmable by Webserver)

*5: If "Send 1s Overload/Underrange" is activated (programmable by Webserver)

<pre> { "sensor": "TA150-<serialnum>-lto", "observations": [{"value": "<S1 value>;<S2 value>;<S3 value>;...;<S60 value>", "timestamp": "<time>"}] }, { "sensor": "TA150-<serialnum>-lto-o", "observations": [{"value": "<S1-o value>;<S2-o value>;<S3-o value>;...;<S60-o value>", "timestamp": "<time>"}] }, { "sensor": "TA150-<serialnum>-lto-u", "observations": [{"value": "<S1-u value>;<S2-u value>;<S3-u value>;...;<S60-u value>", "timestamp": "<time>"}] } </pre>		
	Name	Value description
Optional spectrum values	I006 ... I20k	(*4) 1/3 Octave filter's sound pressure level (see section 1.2)
	I006-o ... I20k-o	(*4) (*5) 1/3 Octave filter's Overload [boolean]
	I006-u ... I20k-u	(*4) (*5) 1/3 Octave filter's Underrange [boolean]

*4: If optional FR150 module is installed and activated (programmable by Webserver)

*5: If "Send 1/3 Overload/Underrange" is activated (programmable by Webserver)

2.3.2 Response data

The server will respond with an HTTP/HTTPS status code "200" if the request was accepted and processed correctly. Other code different to "200", the sensor will consider a server's error.

2.3.3 Retrieving orders

An API must be created on the server with domain *host_name* and located in the path *link_order*. The API makes a GET to the TA150. And then the TA150 reads the Command name and value.

Protocol description:

Format	Protocol	Method
JSON	HTTP, HTTPS MQTT-TCP, MQTT-TLS	GET

Header description:

```
GET /order/<provider_id>/TA150-<serialnum> HTTP/1.1
Host: <host>:<port>
API-KEY: <token>
Content-Type: application/json; charset=UTF-8
```

Parameters	Name	Description
	provider_id	(*) Sentilo provider ID
	serialnum	Device serial number (Txxxxxx)
	host	(*) Server API domain name or IP
	port	(*) Server API port in 4 digits
	token	(*) Server API key secure

*: Programmable by Webserver

If **there is an order**, the server response body will be:

```
{"orders": [{"order": "<cmd_name> <cmd_value>"}]}
```

Parameter	Description
cmd_name	Command name
cmd_value	Command value

The TA150 accepts a lot of prebuilt online commands.

Please see **section 1.3** for those online commands.

2.3.4 Examples

To send data:

- Optional modules: BA150 (battery) - Serial number sensor: T123456 - URL: yourserver.com/sensor/data/ - Port: 80 - Token: abcdefgh12345678 - Integration time: 60s	- "Send Overload/Underrange" deactivated - "Send 1s registers" activated, and "LASmax function" selected - "Send 1/3 registers" activated - "Send 1/3 Overload/Underrange" deactivated
---	--

The sensor data:

```
PUT /data/path/ HTTP/1.1
Host: yourserver.com
IDENTITY_KEY: abcdefgh12345678
Content-Type: application/json; charset=UTF-8
```

```
{ "sensors": [
  {
    "sensor": "TA150-T123456-tl",
    "observations": [{"value": "60", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-la",
    "observations": [{"value": "22.7", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-lc",
    "observations": [{"value": "31.3", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-lz",
    "observations": [{"value": "35.1", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-lai",
    "observations": [{"value": "23.1", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-lafmx",
    "observations": [{"value": "23.3", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-lasmx",
    "observations": [{"value": "23.1", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-laimx",
    "observations": [{"value": "23.6", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-lcpk",
    "observations": [{"value": "45.5", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-lzpk",
    "observations": [{"value": "49.1", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-fls_n",
    "observations": [{"value": "lasmx", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-l1s",
    "observations": [{"value": "22.7;22.7;22.5;22.6;22.7;22.7;22.7;22.9;22.9;22.9;22.8;22.8;22.7;22.9;22.8;22.8;22.9;22.9;22.9;23.0;23.0;22.9;22.8;22.7;22.6;22.6;22.5;22.3;22.3;22.4;22.4;22.3;22.5;22.4;22.4;22.5;22.4;22.5;22.5;22.6;22.7;22.8;22.8;22.9;22.9;23.0;22.9;22.9;22.8;22.8;22.8;22.9;22.9;22.9;22.9;23.0;23.1", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-l1to",
    "observations": [{"value": "23.5;23.3;24.0;24.3;23.8;23.7;23.5;22.9;22.2;27.5;20.5;19.3;18.1;16.4;20.4;12.9;12.2;11.7;9.0;7.3;5.3;7.9;8.4;8.7;6.3;6.0;5.3;5.5;7.3;6.6;15.5;16.9;18.3;14.6;12.6;23.6", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-bat",
    "observations": [{"value": "78", "timestamp": "12/04/2023T11:44:00UTC"}]
  }, {
    "sensor": "TA150-T123456-pwr",
    "observations": [{"value": "3", "timestamp": "12/04/2023T11:44:00UTC"}]
  }
]
```

The server will respond the following data:

```
HTTP/1.1 200 OK
Date: Thu, 12 Apr 2023 11:44:38 GMT
Content-Type: application/json; charset=UTF-8
```

To retrieve orders:

- api_host = yourserver.com
- api_linkorder = order/path/
- api_token = abcdefgh12345678
- id_value = T123456

The data sensor:

```
GET /order/path/TA150-T123456 HTTP/1.1
Host: yourserver.com
IDENTITY_KEY: abcdefgh12345678
Content-Type: application/json; charset=UTF-8
```

When there isn't an order to change, the server will respond the following data:

```
HTTP/1.1 200 OK
Date: Thu, 26 Mar 2020 09:50:41 GMT
Content-Type: application/json; charset=UTF-8
```

And when there is an order (f.e. change the averaging time to **30 seconds**), the server will respond the following data:

```
HTTP/1.1 200 OK
Date: Thu, 26 Mar 2020 09:50:41 GMT
Content-Type: application/json; charset=UTF-8

{"orders":[{"order":"t 0030"}]}
```

3.1 Custom

3.1.1 Sending files

Header description:

```
POST/<audiopath> HTTP/1.1
Host: <audiohost>
IDKEY: <audiokey>
Model: <model>
Serialnum: <serialnum>
Version: <version>
filename: <name>
filetype: <type>
```

Parameters	Name	Description
	host	(*1) (*4) Server API domain name or IP
	path	(*1) (*4) Server API path
	key	(*1) (*4) Server API Key secure
	model	model of the device
	serialnum	Device serial number (Txxxxxx)
	version	Firmware version of the model
	filetype	(*1) (*4) File type: mp3, flac or wav
	filename	File name (serial num. + UTC timestamp: Txxxxxxx_yyyy_mm_dd_hh-ii_ss)

*1: Programmable by Webserver

*8: If optional GA150 module is installed and activated (programmable by Webserver)

Body description:

Audio file content in binary.

3.1.2 Examples

To send data:

- Optional modules: GA150 (Audio) - Serial number sensor: T123456 - URL: http://yourserver.com/sensor/audio/ - Port: 80 - Token: abcdefgh12345678	- Audio type: FLAC - Audio trigger 1 - o Trigger mode: Continuous - o Interval: 24h
---	--

The sensor data:

```
POST /sensor/audio/ HTTP/1.1
Host: yourserver.com
Idkey: abcdefgh12345678
model: TA150
serialnum: T123456
version: 99.00
filename: T259382_2033-04-12_11-43-00.flac
filetype: flac
Content-Length: 1277005
Connection: close

[..audio file content..]
```

The server will respond the following data:

```
HTTP/1.1 200 OK
Date: Thu, 12 Apr 2033 11:44:38 GMT
Content-Type: application/json; charset=UTF-8
```



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Reserves the right to amend the characteristics and accessories in this manual without prior notice.